

LEGAL-TECHNICAL FORENSIC REPORT

Cross-Border ATM & Credit-Card Fraud Ring: Operation Phantom Swipe

Assignment 1 | Unit 1: Cyber Crime Investigation And Digital Forensic

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Classification:	Confidential - LE use	Status:	Early-phase investigation

Executive Summary

This report documents the early-phase digital-forensic investigation of an organised, cross-border payment-fraud ring operating between New Delhi, India, and Dubai, United Arab Emirates. Two exhibits were seized and examined: EXH-01, an ATM magnetic-stripe skimmer with a PIN-pad overlay, and EXH-02, an operator's Android smartphone. Forensic imaging was performed under write-blocking controls, and all 14 acquired files were fingerprinted with SHA-256 to establish an integrity baseline.

A scripted examination recovered 19 artefacts, including harvested card numbers (all Luhn-valid), intercepted OTPs, command-and-control identifiers, cryptocurrency identifiers, and geolocation data tying the suspect device to the crime scenes. A password-protected vault of stolen card "dumps" was lawfully opened by a dictionary attack in 41 guesses (0.0028s), due to the password being weak and reused across the operation. The evidence establishes offences under the Information Technology Act, 2000, the Indian Penal Code, 1860, and conduct addressed by the Budapest Convention. The principal operational obstacle is the cross-border dimension, which this report analyses before recommending concrete improvements to law-enforcement standard operating procedures.

Scope, Authority, and Limitations

- This examination was conducted on two exhibits lawfully seized under a search authority and covers the early phase of the investigation.
- The scope includes the classification of the offences, integrity-preserving acquisition, an initial media search, and decryption of one protected container.

- It does not extend to live network capture, third-party server data (which requires separate legal process), or the tracing of proceeds beyond the wallet identifiers recovered on the devices.
- All findings are reproducible from the accompanying repository.
- As a simulated academic exercise, the dataset is fabricated: card numbers are publicly published network test PANs, and every name, e-mail, phone number, wallet, and coordinate is fictitious. Nothing herein reflects real individuals or accounts.

1. Summary of Crimes and Digital Footprint

1.1 Nature of the Offences

The scenario is a layered criminal enterprise rather than a single offence. Physical ATM skimming captured magnetic-stripe Track-1/Track-2 data together with PINs via an overlay keypad. That data enabled card cloning and online/card-not-present fraud using intercepted one-time passwords. The stolen data ("dumps") was then trafficked to an overseas buyer through a darknet-linked mobile application, with proceeds routed to cryptocurrency wallets. The coordinated roles of installer, mules, and buyer evidence a criminal conspiracy spanning two jurisdictions.

1.2 Digital Footprint Recovered

- **Skimmer (EXH-01):** Includes a capture buffer with six harvested PANs and a PIN keylog; device configuration showing Bluetooth exfiltration to the operator phone; and firmware whose carved strings reveal the C2 host, collector MAC, a Bitcoin wallet, and a `DUMP_KEY` reused as the vault password.
- **Phone (EXH-02):** Includes an SMS backup showing OTP interception; a WhatsApp group coordinating cash-outs across Delhi and Dubai; a sideloaded carding app; a repackaged phishing banking app; an e-mail to the overseas buyer offering "20 fresh dumps"; and photo EXIF metadata plus a location history geolocating the device to the ATM sites and a Dubai cash-pickup point.
- **Protected Vault (EXH-02):** An encrypted archive that yielded a full card-dump CSV, a buyer invoice, and a dummy wallet seed once opened.

1.3 Reconstructed Timeline

Date (2024)	Event	Supporting Artefact
02 Nov	Skimming at HDFC Connaught Place ATM #112; 6 cards cloned; OTP intercepted for online purchase.	<code>captured_tracks.log</code> <code>sms_backup.txt</code> <code>IMG_2041</code> EXIF
03 Nov	Operations moved to Axis Bank Dubai Branch #07; further captures; cash-out coordination.	<code>whatsapp_chat.txt</code> <code>photo_metadata.csv</code>

04 Nov	Dumps packaged in vault.zip; e-mail to Buyer-DXB; crypto payment referenced.	buyer_deal.eml carderpro_config.json
05 Nov	Premises search; EXH-01 and EXH-02 seized and imaged.	Chain-of-Custody form

2. Evidence-Handling Justification

Digital evidence is volatile and trivially altered; its evidentiary value depends entirely on demonstrable integrity and an unbroken chain of custody.

2.1 Seizure and Isolation

The exhibits were photographed in situ before removal. EXH-02 (the phone) was immediately placed in a Faraday/RF-shielded bag to sever cellular, Wi-Fi, and Bluetooth connectivity, defeating remote-wipe and remote-lock commands that could destroy evidence. EXH-01 was bagged anti-static with its power state recorded, respecting the order of volatility.

2.2 Write-Blocking and Imaging

All analysis was performed on working copies acquired through a hardware write-blocker; original media were never mounted read-write. This guarantees that examination cannot alter source evidence.

2.3 Hashing and Integrity (Why SHA-256)

Each acquired file was hashed with SHA-256 at acquisition, producing a verifiable manifest of 14 entries (`hashes/SHA256SUMS.txt`). A cryptographic hash acts as a tamper-evident seal; re-hashing before analysis and submission proves the evidence is unchanged. SHA-256 is used over legacy MD5/SHA-1 because it is not collision-weak.

2.4 Admissibility of Electronic Records

Under Indian law, electronic records are admissible subject to a certificate under Section 65B of the Indian Evidence Act, 1872 (now Section 63 of the Bharatiya Sakshya Adhiniyam, 2023). The write-blocked acquisition, contemporaneous hashing, and chain of custody support this certificate.

2.5 Chain of Custody

Every transfer is logged with identities, timestamps, and purposes in `Chain_of_Custody_Form.pdf` . Tamper-evident seals were applied and recorded to detect any interference.

3. Forensic Findings and Cryptography

3.1 Artefact Recovery

A scripted string-search, metadata, and e-mail examination catalogued 19 artefacts. All recovered card numbers pass the Luhn checksum, confirming they are well-formed PANs. Geolocation

corroborates the SMS/WhatsApp timeline across both countries, and e-mail headers provide network-attribution leads.

Artefact Category	Examples	Investigative Value
Payment card numbers	5 unique, all Luhn-valid	Victim identification; fraud-loss quantum
Intercepted OTPs/SMS	OTP 448190, 771020	Proves CNP fraud method
Geolocation	Delhi, Dubai, pickup point	Places device at scenes/times
C2/darknet + IP	phantom-swipe.onion 185.220.101.47	Infrastructure attribution
Crypto wallets	BTC + USDT-TRC20	Money-trail / proceeds tracing
Correspondents	buyer/operator e-mails	Links ring members across borders

3.2 Cryptography Component

The stolen dumps were sealed in a password-protected archive (ZipCrypto). A dictionary attack recovered the password `cashout2024` in 41 guesses in 0.0028 seconds. The archive fell quickly because ZipCrypto is a weak legacy cipher, and the password was short, lowercase, and reused across the operation.

3.3 Ethics: Brute-Forcing vs. Lawful Decryption

Brute-force decryption by investigators is lawful only within a valid search warrant or seizure authority and must be limited to seized evidence and fully logged. In India, Section 69 of the IT Act empowers directions for decryption assistance, though compelling a suspect raises self-incrimination issues (Article 20(3) of the Constitution).

3.4 Reflection on Password Strength

This case illustrates a recurring reality: offenders reuse short, memorable passwords, making their operational security their undoing. Strong, unique, high-entropy passphrases and modern authenticated encryption (e.g., AES-256) resist dictionary attacks.

4. Challenges in Cross-Border Investigation

The India-UAE dimension creates several structural obstacles:

- **No Budapest Convention Route:** India is not a party to the Convention on Cybercrime, so its expedited mechanisms are unavailable, and cooperation relies on MLATs and Letters Rogatory, which take months.
- **Foreign-Held Data:** Chat and e-mail records sit with providers outside Indian jurisdiction, requiring specific legal processes.
- **Attribution Behind Anonymity:** Tor C2 and disposable identifiers frustrate direct attribution.
- **Cryptocurrency Tracing:** Following proceeds demands blockchain analytics and VASP cooperation, engaging the FATF Travel Rule.
- **Dual Criminality:** MLA typically requires the conduct to be an offence in both states.
- **Time and Volatility:** Cross-border latency clashes with short data-retention windows.

5. Recommendations for Law-Enforcement SOPs

- **First-Responder Kit & Checklist:** Mandate Faraday bags, hardware write-blockers, and a seizure checklist.
- **Hashing + Certificate Templates:** Require SHA-256 hashing at acquisition and attach a pre-formatted Section 63 BSA/65B certificate.
- **Rapid Preservation Playbook:** Maintain templates for major foreign providers and a roster of MLA points of contact.
- **Dedicated Cyber-MLA Cell:** Resource a specialised unit to draft and track MLAT requests.
- **Crypto-Tracing Capability:** Adopt blockchain-analytics tooling and formal VASP liaison channels.
- **Bank/ATM Partnership:** Establish an anti-skimming programme with issuers and operators.
- **Training & Audit:** Certify examiners, validate tools, and keep tamper-evident audit logs.
- **Joint Task Forces:** Use INTERPOL and joint task forces to coordinate simultaneous action.

6. Conclusion

Operation Phantom Swipe demonstrates how foundational digital-forensic discipline converts seized hardware into court-ready evidence of a multi-jurisdiction fraud enterprise. The technical investigation is tractable; the binding constraint is cross-border cooperation. Closing that gap through faster preservation, MLA capacity, and VASP partnerships is where the greatest investigative return lies.

Appendices

Appendix A: Condensed Statutory Mapping

Conduct	IT Act 2000	IPC 1860	Budapest
ATM skimming/interception	s.66 r/w s.43	s.378/379	Art. 2, 3
Card cloning (forgery)	s.66	s.463/465/468/471	Art. 4, 7
Identity theft (PAN/PIN/OTP)	s.66C	s.419	Art. 8
Online / CNP fraud, phishing app	s.66D	s.420	Art. 8
Trafficking dumps/device misuse	s.66B	s.411	Art. 6
Organised ring		s.120B r/w s.34	

Appendix B: Tools Used

Python 3 (hashlib , zipfile , re , csv); coreutils sha256sum ; Info-ZIP zip/unzip ; John the Ripper/hashcat; ReportLab. Custom scripts: generate_evidence.py , hash_files.py , search_media.py , crack_zip.py , seal_vault.sh , crack_with_john.sh .

Appendix C: Authorship Declaration

I, Kartik Kumar, declare that this report and the accompanying repository are my own work, produced for Cyber Crime Investigation And Digital Forensic at K.R. Mangalam University. All case data is simulated for academic purposes; card numbers are published network test values and no real personal or financial data was used. Sources and tools are acknowledged above.

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